MTH1001-B

UNIVERSITY OF EXETER

FACULTY OF ENVIRONMENT, SCIENCE AND ECONOMY

MATHEMATICS AND STATISTICS

May 2024

Mathematical Structures

Module Leader: Dr Stephen Mc Guire

Duration: 2 HOURS

Answer Section A (50%) **and any TWO of the three questions in Section B** (25% **for each).**

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

SECTION A

1. (a) Using the Euclidean Algorithm, calculate the $\gcd(12378, 3054)$ and express it in the form $12378a + 3054b = \gcd(12378, 3054)$ for suitable integers a and b. Hence, find an integer x in the range $0 \leq x < 3054$ such that $12378x \equiv 192 \pmod{3054}$. (12)

   (b) Let $\sigma = (137)(56)$ and $\tau = (123)(489)$ be elements of the symmetric group S_9. Express each of the permutations $\sigma\tau$, $\tau\sigma$ and $\tau\tau^{-1}\sigma^{-1}\tau\sigma$ as a product of disjoint cycles. Find the order and the sign of each of these permutations. (13)

   (c) Consider the vectors

$$\overrightarrow{v_1} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}, \quad \overrightarrow{v_2} = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 3 \end{pmatrix}, \quad \overrightarrow{v_3} = \begin{pmatrix} 2 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad \overrightarrow{v_4} = \begin{pmatrix} 3 \\ 1 \\ 2 \\ 2 \end{pmatrix}$$

   and

$$\overrightarrow{b} = \begin{pmatrix} 3 \\ 4 \\ 7 \\ 11 \end{pmatrix}.$$

   (i) Find a maximal linearly independent sublist of the list of vectors $\overrightarrow{v_1}$, $\overrightarrow{v_2}$, $\overrightarrow{v_3}$, $\overrightarrow{v_4}$.

   (ii) Determine whether $\overrightarrow{b}$ can be written as a linear combination of $\overrightarrow{v_1}$, $\overrightarrow{v_2}$, $\overrightarrow{v_3}$, $\overrightarrow{v_4}$ and, if it can, write it in this way. (13)

   (d) Let V be a real vector space. Suppose $\overrightarrow{v_1}, \ldots, \overrightarrow{v_n}$ are linearly independent vectors of V. Show that

$$\lambda_1\overrightarrow{v_1}, \ldots, \lambda_n\overrightarrow{v_n}$$

   are also linearly independent vectors for any $\lambda_1, \ldots, \lambda_n \in \mathbb{R} \setminus \{0\}$. (12)

[50]

SECTION B

2. (a) State the Fundamental Theorem of Arithmetic. (2)

   (b) Prove that $\sqrt{29}$ is irrational. If you use the Fundamental Theorem of Arithmetic in your proof, you should indicate clearly where you do so. (12)

   (c) Determine whether each of the following statements is true or false, and give a proof (if it is true) or a counterexample (if it is false).

   (i) Let G be a group, and let $g, h, k \in G$. If $ghg^{-1} = gkg^{-1}$, then $h = k$.

   (ii) Suppose that a group consists of three distinct elements $\{x, y, z\}$ and that x is the identity. Then this group must be abelian. (11)

   [25]

3. (a) (i) Let $\theta : G_1 \to G_2$ be a map between two groups G_1 and G_2. Define what it means to say that the map θ is a group homomorphism. (2)

   (ii) Let $\theta : G_1 \to G_2$ be a group homomorphism. Prove that $\theta(e_1) = e_2$, where e_1 and e_2 are the identity elements in G_1 and G_2, respectively.
   Hence, or otherwise, prove that $\theta(g^{-1}) = (\theta(g))^{-1}$ for all $g \in G_1$. (8)

   (iii) Let $a, b \in \mathbb{R}^\times$ and consider the map

   $$\theta : \mathbb{Z}^2 \to \mathbb{R}^\times, \quad \theta(m, n) = a^m b^n.$$

   Determine whether or not θ is a group homomorphism, making sure to justify your answer. (6)

   (b) (i) Let U and V be real vector spaces. What does it mean for T to be a linear map from U to V? (2)

   (ii) Let A be a $m \times n$ real matrix, and let

   $$F_A : \mathbb{R}^n \longrightarrow \mathbb{R}^m,$$

   be the map defined by

   $$F_A(\overrightarrow{x}) = A\overrightarrow{x},$$

   for any $\overrightarrow{x} \in \mathbb{R}^n$. Show that F_A is linear. (7)

   [25]

4. (a) Let $a, b, c, d, n \in \mathbb{Z}$ with $n \geq 1$.

(i) Prove that if $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$ then $ac \equiv bd \pmod{n}$. (4)

(ii) Using part (i), or otherwise, prove that if $a \equiv b \pmod{n}$ then $a^k \equiv b^k \pmod{n}$ for all $k \in \mathbb{N}$. (6)

(iii) Is it also true that if $a^k \equiv b^k \pmod{n}$ for some $k \in \mathbb{N}$ then $a \equiv b \pmod{n}$?
Justify your answer with either a proof or a counterexample. (3)

(b) Let V be a real vector space of dimension 3 and let $\{\overrightarrow{v_1}, \overrightarrow{v_2}, \overrightarrow{v_3}\}$ be a basis for V. Consider the two sets

$$U = Span\{\overrightarrow{v_1} + \overrightarrow{v_2}, \overrightarrow{v_1} - \overrightarrow{v_2}\}, \quad W = Span\{\overrightarrow{v_2} + \overrightarrow{v_3}, \overrightarrow{v_2} - \overrightarrow{v_3}\}.$$

(i) Show that U and W are both subspaces of V;

(ii) Define

$$U + W = \{\overrightarrow{u} + \overrightarrow{w} : \overrightarrow{u} \in U, \overrightarrow{w} \in W\}.$$

Show that

$$V = U + W.$$

(12)

[**25**]